

CURRICULUM VITAE

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SUMMARY

Mr. Samuel Aklashie is a Graduate Teaching and Research Assistant at the Department of Construction Technology and Management, KNUST, Ghana. He holds a BSc. in Construction Technology and Management and currently completed an MPhil. in Building Technology at KNUST. His research focuses on sustainable construction, circularity, digital product passports, and the integration of AI in the built environment.

He has co-authored multiple high-quality peer-reviewed journal articles and conference papers. He serves on the Editorial Review Board of Construction Innovation and actively reviews for journals including Engineering, Construction and Architectural Management (ECAM) and Journal of Engineering, Design and Technology. In his teaching role, he assists in course delivery, supervises student projects, and supports academic research and departmental activities.

Samuel is proficient in ArchiCAD, Python, Ontology Engineering and project management tools. He is an Incorporate Member of the Ghana Institute of Construction and a UPG Sustainability Leader. He is a self-driven, research-oriented professional passionate about advancing innovation in construction and sustainability.

EDUCATION

2015 – 2018

Building Construction (WAEC Certificate) – *Obrachire Senior High Technical School*

2019 – 2023

BSc. Construction Technology and Management (First Class Honors) – *KNUST, Kumasi, Ghana*

(GPA=3.78/4.00) ([Link](#))

KNUST: 3rd in Ghana; CTM Department: 1st in Ghana

- Thesis: *Implementing Materials Passport in the Construction Industry: Empirical Evidence from Ghana*.
- Supervised by Prof. Kofi Agyekum; later developed into a journal publication ([Link](#)).

2025 – 2026

MPhil Building Technology – *Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana* (GPA=3.83/4.00) ([Link](#)).

- Thesis: *A Structural Model of Design for Greater Efficiency Implementation Among Design Professionals in the Ghanaian Construction Industry* ([Link](#)).
- Methodology: Questionnaire survey of 238 professionals (architects, civil/structural engineers, building services engineers); data analyzed with descriptive statistics and structural equation modelling (SEM).
- Key Focus: Awareness, implementation behavior, challenges, and strategies for *DfGE*.
- Supervisors: Prof. Kofi Agyekum & Prof. Emmanuel Adinyira.
- Partially Funded by a Research Grant, Brew-Hammond Energy Centre, KNUST.

RESEARCH PUBLICATIONS

JOURNAL ARTICLES

1. **Aklashie, S., Agyekum, K., Tchao, E., Adinyira, E., & Nani, G. (2026)** “Blockchain-Digital Twin Integration in Construction Material Passports: Preparing for Digital Product Passport Implementation”. *Built Environment Project and Asset Management*. (Corresponding Author) <https://doi.org/10.1108/BEPAM-07-2025-0188>
 - Type of Publication: Journal Article (Refereed journal)
 - Authorship Statement: Conceptualization, protocol design, database search/screening, quality appraisal, evidence synthesis, drafting, and submission/revision management.
2. **Aklashie, S., Agyekum, K., Adinyira, E., & Chileshe, N. (2026)**, “Socio-Technical Implementation Dynamics in Carbon Reduction Awareness–Practice Gaps”, *Engineering, Construction and Architectural Management*. <https://doi.org/10.1108/ECAM-08-2025-1280>
 - Type of Publication: Journal Article (Refereed journal)
 - Authorship Statement: I led conceptual framing, study design, analytic strategy, manuscript drafting, and coordinated co-author responses to editorial feedback.
3. **Aklashie, S., Agyekum, K., Adinyira, E., Gyimah, K.A., Opoku, D., & Twumasi, J.O. (2025)**, “Design for Greater Efficiency Awareness: A Critical Assessment of Ghana’s Design

Professionals”. *Built Environment Project and Asset Management*, <https://doi.org/10.1108/BEPAM-10-2025-0337>

- Type of Publication: Journal Article (Refereed journal)
 - Authorship Statement: S. Aklashie led the conceptualization, study design, data collection, analysis, manuscript drafting, and submission coordination.
4. Alex, O., Danquah, R. D., **Aklashie, S.**, Larbi J. A., Qiao, Y. (2026), “Digital green finance models for sustainable urban development: Current trends and future perspectives” [J]. *Green Finance*, 8(1): <https://doi.org/10.3934/GF.2026003>
- Type: Journal Article (Refereed).
 - Authorship Statement: Conceptualization, Investigation, Methodology, Formal analysis, Writing, Reviewing, and Editing of the research paper.
5. Dompey, A.M.A., **Aklashie, S.**, Danso, F.O., Agyekum, K., Gasue, R., and Kumi-Amoah E. (2025). “Digital innovation in building maintenance: the perspectives of facilities managers in Ghana,” *International Journal of Building Pathology and Adaptation*. (**Corresponding Author**) <https://doi.org/10.1108/IJBPA-01-2025-0009>
- Type: Journal Article (Refereed).
 - Authorship Statement: Conceptualization, data collection, analysis, and manuscript drafting.
6. Pitti, H., **Aklashie, S.**, Rathnayake, A. G. G., Sackey, N. Y. A., Agyekum, K. and Sackey, F. N. (2025). “The Role of Internet of Things in Managing Carbon Emissions in the Construction Industry: A Systematic Review,” *Intelligent Infrastructure and Construction*. <https://doi.org/10.3390/iic1030008>
- Type: Journal Article (Refereed).
 - Authorship Statement: I contributed to methodology, and original draft preparation.
7. Agyekum, K., Botchway, S.Y., Dompey, A.M.A., Gasue, R. and **Aklashie, S.** (2025). “Economic Performance Indicators for Assessing the Sustainability of Construction Projects: Stakeholders’ Perspectives.” *Built Environment Project and Asset Management*, pp.1–18. <https://doi.org/10.1108/BEPAM-09-2024-0221>
- Type: Journal Article (Refereed).
 - Authorship Statement: I contributed to data analysis, presentation of results and manuscript review.
8. Kotei-Martin, J.N., Agyekum, K., Salgin, B., Pittri, H., Botchway, E.A., Gasue, R., & **Aklashie, S.** (2025), “Challenges to the implementation of design for adaptability among

design professionals in Ghana”, *International Journal of Building Pathology and Adaptation*. Advance online publication. <https://doi.org/10.1108/IJBPA-09-2024-0189>

- Type: Journal Article (Refereed).
 - Authorship Statement: Data collection, literature review, and manuscript drafting.
9. Gasue, R., **Aklashie, S.**, Dompey, A.M.A., Agyekum, K., & Opoku, D. (2024), “Implementing materials passports in the construction industry: Empirical evidence from Ghana”, *International Journal of Building Pathology and Adaptation*, 42(6), 1201–1223. <https://doi.org/10.1108/IJBPA-01-2024-0007>
- Type: Journal Article (Refereed).
 - Authorship Statement: I wrote the introduction statement, co-designed the study, conducted data analysis, and drafting of other sections of the manuscript.

CONFERENCE PAPERS

1. Agyekum, K., **Aklashie, S.**, Acquaaah Agyekum, M., Tchao, E.T., & Addo, P.K. (2025), “The role of generative AI in built environment research: A systematic literature review” Paper presented at the E-Learning International Conference, Kumasi, Ghana. ([Link](#))
 - Type: Conference Paper (Refereed)
 - Authorship Statement: I conducted the systematic review and presented at the conference ([Link](#)).
2. Botchway, B., Adinyira, E., **Aklashie, S.**, Asamoah-Tieku, F., & Apeagyei, E.Y. (2024), “Reviewing the culture of health and safety management in the Ghanaian construction industry”, Paper presented at the CIB International Conference, Accra, Ghana. ([Link](#))
 - Type: Conference Paper (Refereed).
 - Authorship Statement: I contributed to literature review, co-drafted the manuscript and presented at the conference.
3. Botchway, B., Ghansah, F.A., **Aklashie, S.**, Ackah, B., Kove, D., & Ohene Adjei, T. (2024), “Impact of sustainable buildings on housing development in Ghana”, Paper presented at the 1st International Conference on Environment, Social, Governance and Sustainable Development of Africa (ICESDA). ([Link](#))
 - Type: Conference Paper (Refereed)
 - Authorship Statement: I participated in data analysis and preparation of the presentation.

BOOK CHAPTER PUBLICATIONS

1. Agyekum, K., **Aklashie, S.**, Opoku, A., Salgin, B., & Gasue, R. (2025), “Circularity and Building Systems for Green Buildings”, In The Elgar Companion on Circularity, Deconstruction, and Adaptability in Green Buildings. Edward Elgar Publishing. (Link)
 - Type: Book Chapter (Refereed)
 - Authorship Statement: I undertook the majority of the writing and drafting.
2. Agyekum, K., Amudjie, J., **Aklashie, S.**, Salgin, B., Opoku, A., & Gasue, R. (2025), “Circular construction practices for green buildings”, In the Elgar Companion on Circularity, Deconstruction, and Adaptability in Green Buildings. Edward Elgar Publishing (Link)
 - Type: Book Chapter (Refereed)
 - Authorship Statement: Contributed to literature review and analysis of case examples.
3. Dompey, A.M.A, **Aklashie, S.**, and Kofi Agyekums (2025). “User-centric Design Considerations for Green Buildings.” In the Elgar Companion on Circularity, Deconstruction, and Adaptability in Green Buildings. Edward Elgar Publishing.
 - Type: Book Chapter (Refereed)
 - Authorship Statement: Contributed to literature review and analysis of case examples.

JOURNAL ARTICLES (UNDER REVIEW)

1. Bassily, V. E., **Aklashie, S.**, Sherif G. and Opoku, A. (2026), “Health and Well-Being Benefits of Nature-Based Solutions for Climate Mitigation and Adaptation in Middle East and African Urban Environments: A Review”. *Nature-Based Solutions*.
 - Type of Publication: Journal Article (Refereed journal)
 - Authorship Statement: study design, analysis and manuscript drafting
2. **Aklashie, S.**, Agyekum, K., Gasu, Pittri, H., and Adinyira, E., (2026) “Design for Greater Efficiency in Practice: Implementation Patterns and Capacity Gaps Among Ghana’s Design Professionals”. *Journal of Engineering, Design and Technology*. (Corresponding Author)
 - Type of Publication: Journal Article (Refereed journal)
 - Authorship Statement: Conceptualization, protocol design, database search/screening, quality appraisal, evidence synthesis, drafting, and submission/revision management.

AWARDS, GRANTS & SCHOLARSHIPS

Academic Excellence & Recognition

- Overall Best Published Paper Award, 3rd Postgraduate Research Conference (PGRC 2026) awarded for “*Socio-Technical Implementation Dynamics in Carbon Reduction Awareness-Practice Gaps*” ([Link](#))
- First Class Honors, BSc Construction Technology and Management, KNUST (2023)

Research Grants & Funding

- MPhil Research on Green Building, The Brew-Hammond Energy Centre, College of Engineering, KNUST ([Link](#))

Scholarships & Fellowships

- MasterCard Foundation Scholar, KNUST (Sep 2019 – Nov 2023) ([Link](#))
[Leadership, strategy, and project management development]

Leadership & Professional Recognition

- UPG Sustainability Leader, United People’s Global (Jan 2022 – Present) ([Link](#))
[Community education on sustainability; collaborative projects]

EDITORIAL BOARD MEMBERSHIP

- Member, Editorial Review Board, *Construction Innovation* (Emerald Publishing, Q1): Aug 2026–Present

PEER-REVIEW CONTRIBUTIONS (54) ([Link](#))

- Journal of Engineering, Design and Technology: **29** reviews; Engineering, Construction and Architectural Management: **8** reviews; PLOS ONE: **6** reviews; Construction Innovation: **5** reviews; International Journal of Building Pathology and Adaptation: **5** reviews; Journal of Facilities Management: **1** review

RESEARCH EXPERIENCE

Research Assistant / Collaborator

Building Science, Engineering and Materials Research Team, Department of Construction Technology and Management, KNUST – Kumasi, Ghana
Nov 2023 – Present

- Actively engaged in research on sustainable construction, green building, and digital innovation within the Ghanaian built environment.

- Assisted Prof. Kofi Agyekum (Editor) in the development of *The Elgar Companion on Circularity, Deconstruction, and Adaptability in Green Buildings* (Edward Elgar Publishing). Played a key role in:
 - Reviewing submitted chapters for quality and coherence.
 - Synthesizing abstracts of all chapters into a unified Introductory Chapter.
- Worked directly with the Scientific Committee of the 2025 CIB Ghana Conference, coordinating authors' submissions, reviewing content for compliance with academic standards, and supporting the organization of peer review. This role enhanced my skills in academic coordination, editorial review, and scholarly communication.
- **Skills and Competencies**
- **Research Skills:**
 - Systematic Literature Reviews (SLR) and Meta-Analyses.
 - Ontology engineering
 - Data analysis using structural equation modelling (SEM) and SPSS
 - Programming with Python for research-related computational tasks.

PUBLIC OUTREACH & COMMENTARY

1. Adinyira, E. and **Aklashie, S.** (2025, June 10). Shrinking to increase: The economic and structural risks with decreasing building material sizes. MyJoyOnline.
 - Acknowledgement: Prof. Emmanuel Adinyira credited me for supporting research contributions.
 - [Read Online](#)
2. Adinyira, E. and **Aklashie, S.** (2024, Feb 12). What the 24-Hour Economy and Digitalisation Could Mean for the Ghanaian Construction Industry (Part 2). MyJoyOnline.
 - Acknowledgement: Prof. Emmanuel Adinyira formally acknowledged me for providing research support as Teaching and Research Assistant. [Read Online](#)

PROFESSIONAL EXPERIENCE

Graduate Teaching and Research Assistant

Department of Construction Technology and Management, KNUST – Kumasi, Ghana
Nov 2023 – Present

Responsibilities include:

- Organizing and delivering tutorial classes in Building Science, Strength of Materials, and Research Methods.

- Assisting supervisors in updating lecture notes, preparing teaching materials, and coordinating academic content.
- Supporting departmental research activities: designing questionnaires, conducting surveys, reviewing literature, and preparing manuscripts for seminars and conferences.
- Assisting in supervision of over 25 undergraduate and 10 postgraduate students' research projects, ensuring quality of data collection, analysis, and reporting.
- Performing administrative duties, including seminar coordination and departmental support.

Intern – Antartic LTD

Oct 2022 – Dec 2022

- Fostered collaboration by supporting digital communication tools and enhancing stakeholder engagement.
- Promoted efficient workflows through digital project management platforms.
- Contributed to team-based construction project coordination.

Intern – Conad's Estate LTD

Oct 2021 – Dec 2021

- Maintained accurate departmental records using Google Workspace.
- Analyzed engagement data with Google Analytics to support digital marketing strategies and improve visibility.
- Prepared visual content and reports for internal communication and outreach.

PROFESSIONAL SERVICE AND LEADERSHIP

- **Rapporteur:** *E-Learning Conference* (07/2025)
Documented proceedings, prepared detailed session reports, and contributed to final conference summaries.
- **Logistics Committee Member & IT Support Team Member:** *CIB Conference, Ghana* (10/2024)
Oversaw event logistics, coordinated transportation, and provided IT/AV support to ensure smooth conference operations.
- **Evaluator:** *UPG Sustainability Selection Council* (04/2024) ([Link](#))
Evaluated applications for the UPG Sustainability Leadership program.

PROFESSIONAL MEMBERSHIPS

- Ghana Institute of Construction: Incorporate Member.

CERTIFICATIONS

- EDGE Experts training (2025) ([Link](#))
- Workshop on Systematic Reviews and Meta-analyses: ACSRMs KNUST (2024) ([Link](#))
- Microsoft Products (Udemy): 2022
- Project Management: Centre for Capacity Building & Innovations, KNUST (2021)
- Capacity Building, Coaching & Communication: International Leadership Foundation (2020)
- Character Formation & Integrity; Community Building: International Leadership Foundation (2019)

REFERENCES

- Prof. Emmanuel Adinyira: eadinyira.cap@knust.edu.gh
- Prof. Kofi Agyekum: kagyekum.cap@knust.edu.gh
- Dr. Ernest Kissi: ernestkissi@knust.edu.gh